Inclusion–exclusion, derangements, and restricted assignments

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Fix the universe and properties

Let finite U have size N, with properties $A_1, \ldots, A_n \subseteq U$. An intersection over J means satisfying at least all properties in J; properties outside J may also hold. Define $S_0 = N$ and

$$S_k = \sum_{\substack{J \subseteq [n] \\ |J| = k}} \left| \bigcap_{i \in J} A_i \right|.$$

This sums intersection sizes, rather than measuring their union. An object satisfying c properties appears $\binom{c}{k}$ times in S_k. This multiplicity explains alternating signs, exact-hit formulas, and truncation errors.

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General proof by indicators

Let I_i(x) indicate membership in A_i. The indicator of no hit is $\prod_i (1 - I_i(x))$. Expand and sum over U:

$$\left| U \setminus \bigcup_i A_i \right| = \sum_{J \subseteq [n]} (-1)^{|J|} \left| \bigcap_{i \in J} A_i \right| = \sum_{k=0}^{n} (-1)^k S_k.$$

The empty intersection is U, supplying N. Finite sums commute and each indicator product counts its intersection. Neither independence nor equal intersection sizes is required. Subtracting from N gives union size $S_1 - S_2 + S_3 - \dots$. At n=0, the union is empty and the no-hit count N, consistent with empty product one.

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Why each union member remains once

For c≥1 hits, net multiplicity is

$$\sum_{k=1}^{c}(-1)^{k+1}\binom{c}{k} = 1-(1-1)^{c} = 1.$$

Objects with c=0 enter no positive-order intersection and contribute zero. Four hits contribute 4−6+4−1=1; stopping after pair intersections leaves −2. Multiple overlaps require the full accounting. Individual set sizes alone do not determine intersections; inclusion–exclusion organizes known data, rather than supplying missing data.

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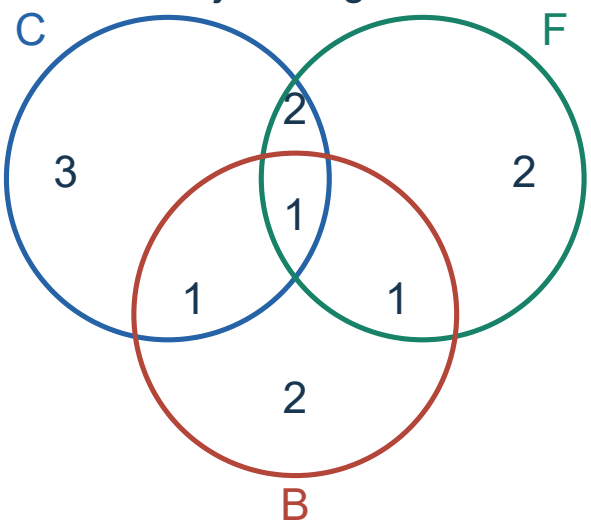
Every region for three defects

Among 20 nodes, defects C,F,B have sizes 7,6,5; pair intersections 3,2,2; triple intersection 1. Pair intersections include triple members.

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Every region for three defects (continued)

Numbers label disjoint regions, not entire intersections



No hits: 8

$N=20$; $|C|=7$ · $|F|=6$ · $|B|=5$

$|C \cap F|=3$ · $|C \cap B|=|F \cap B|=2$

$S_1=18$ · $S_2=7$ · $S_3=1$

At least one: $18-7+1=12$

No hits: $20-12=8$

Exactly two defects give 2,1,1. Only C gives $7-3-2+1=3$; only F gives 2 and only B gives 2. Interior regions total $3+2+2+2+1+1+1=12$, leaving 8 compliant nodes. Equivalently $S_1=18, S_2=7, S_3=1$ and $20-18+7-1=8$. A negative disjoint region would expose inconsistent input intersections, even if an alternating arithmetic expression could still be evaluated.

Truncated alternating binomial sums

For integers $c \geq 1$,$r \geq 0$, with out-of-range binomials zero,

$$\sum_{k=0}^{r} (-1)^k \binom{c}{k} = (-1)^r \binom{c-1}{r}.$$

Substitute Pascal's identity $\binom{c}{k} = \binom{c-1}{k} + \binom{c-1}{k-1}$. Shift the second sum by one; internal terms cancel, leaving the final term. For $r \geq c$ both sides vanish.

Thus an object with $c \geq 1$ hits contributes $1 - (-1)^r \binom{c-1}{r}$ to the union sum through order r. It is at least one for odd r and at most one for even r; outside objects always contribute zero. This proves the direction rather than guessing from signs.

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Bonferroni bounds

For $T_r = \sum_{k=1}^{r} (-1)^{k+1} S_k$, 1≤r≤n, summing the pointwise inequalities gives

$$r \text{ odd}: \quad \left|\cup A_i\right| \le T_r, \qquad r \text{ even}: \quad \left|\cup A_i\right| \ge T_r.$$

No-hit bounds reverse on subtracting from N. These bounds require consistent intersection data through order r and do not require independence. In the defect example, T_1=18 is an upper bound and T_2=11 a lower bound. The union lies in [11,18], compliance in [2,9]. Order three gives exact values 12,8. Truncation is already exact when every higher-order intersection is empty.

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Probability and weights use linearity

Taking expectations of the same finite indicator identity yields

$$P(\cup_i A_i) = \sum_{\varnothing \neq J \subseteq [n]} (-1)^{|J|+1} P(\cap_{i \in J} A_i).$$

Intersection probabilities need not factor. Mutual independence permits that substitution; pairwise independence alone does not give all higher-order products.

Replace cardinality by $\sum_{x \in A} w(x)$. For nonnegative weights the identity and Bonferroni directions survive by multiplying pointwise inequalities and summing. Signed weights retain the algebraic identity but need not retain inequality directions.

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Divisibility intersections use lcm

In 1,…,M, let A_d contain multiples of d. Then $A_a \cap A_b = A_{\operatorname{lcm}(a,b)}$, of size $\lfloor M / \operatorname{lcm}(a,b) \rfloor$. Only coprime divisors permit replacing lcm by product.

For M=40 avoiding 2,3,5, single counts are 20,13,8; pair counts 6,4,2; triple 1. Thus

$$40 - (20 + 13 + 8) + (6 + 4 + 2) - 1 = 10.$$

The numbers are 1,7,11,13,17,19,23,29,31,37. Multiples of both 4 and 6 are multiples of 12, not 24. Alternating signs cannot correct a wrongly modeled intersection.

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Euler's totient product from inclusion–exclusion

The totient ϕ(m) counts integers in 1,…,m coprime to m. Let distinct prime divisors be p_1,…,p_s. Noncoprimality is equivalent to divisibility by at least one of them, since any common divisor above one has a prime divisor. Every subset product d divides m, so its intersection has exactly m/d members, without floor error. Expanding gives

$$\varphi(m) = m \prod_{p \mid m} (1 - 1/p).$$

For m=30, 30(1/2)(2/3)(4/5)=8, with 1,7,11,13,17,19,23,29. At m=1 the empty product gives ϕ(1)=1. For a general upper endpoint M, retain floors; this product cannot be applied unconditionally.

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Bounded nonnegative integer solutions

For m≥1, N,b_i nonnegative integers, count $\sum_i x_i = N, 0 \le x_i \le b_i$. Without upper bounds there are $\binom{N+m-1}{m-1}$ solutions. If every i∈J violates its bound, subtract b_i+1 from x_i. This is a bijection with nonnegative solutions of the reduced total, giving

$$\# = \sum_{J \subseteq [m]} (-1)^{|J|} \binom{N - \sum_{i \in J}(b_i + 1) + m - 1}{m - 1}.$$

A negative remaining total means zero solutions; do not interpret it through generalized negative-upper binomials. For $x_1+x_2+x_3=6$ with bounds 2,3,4, the unrestricted count is 28. Single violations leave totals 3,2,1, giving 10,6,3. Every double violation needs more than six. Thus 28−10−6−3=9.

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Surjections exclude missed outputs

There are n^m functions from m points to n points. Let A_i mean output i is missed. Missing a specified k outputs leaves (n−k)^m functions, so

$$\operatorname{Surj}(m,n)=\sum_{k=0}^{n}(-1)^k\binom{n}{k}(n-k)^m=n!S(m,n).$$

The partition expression counts nonempty fibers, and the alternating expression follows from the general proof. For m<n the count is zero; m=n gives n!; m=0 retains the unique-empty-function convention. Five labeled inputs onto three named outputs give 243−3 • 32+3 • 1−0=150. Exactly two outputs instead requires choosing them first: $\binom{3}{2}(2^5-2)=90$. Deleting a term of the onto-three formula does not change its meaning correctly.

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Circular arrangements require valid blocks

For m≥2 disjoint specified pairs among 2m people, identify rotations but distinguish reflections. If k specified pairs are adjacent, contract each pair into a labeled block. There remain 2m−k distinct objects around the circle and two orders inside each block, giving $2^k(2m-k-1)!$. Hence no specified pair adjacent gives

$$\sum_{k=0}^{m}(-1)^k\binom{m}{k}2^k(2m-k-1)!.$$

At m=4 the terms are 5040−5760+2880−768+96=1488. The condition 2m≥4 excludes the two-person circular degeneracy in which two orientations represent one rotation class. If bad pairs share people, independent two-person contractions are invalid.

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No isolated vertices need not mean connected

There are $2^{\binom{n}{2}}$ labeled undirected simple graphs. If specified k vertices must be isolated, only edges among the other n−k vertices remain free, giving $2^{\binom{n-k}{2}}$. Thus the count without isolated vertices is

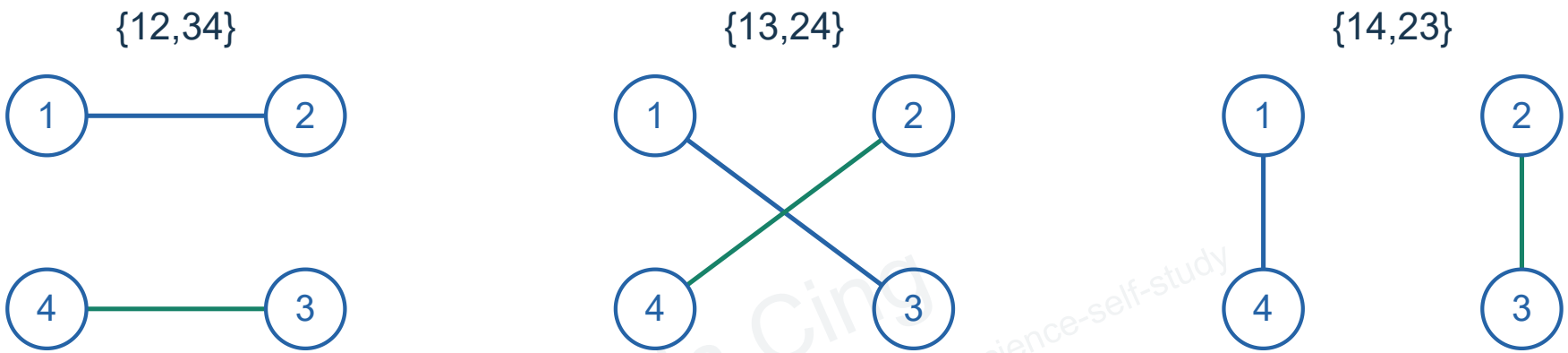
$$\sum_{k=0}^{n}(-1)^k\binom{n}{k}2^{\binom{n-k}{2}}.$$

For n=4 this is 64−32+12−4+1=41. Three are disconnected perfect matchings.

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No isolated vertices need not mean connected (continued)

Four labeled vertices: exactly three disconnected graphs without isolates



All degrees are 1; crossings are not vertices. Connected graphs: 41−3=38.

Every disconnected no-isolate graph on four vertices has two components of size two, each with its unique edge. These three exhaust the cases, so connected graphs number 41−3=38. Inclusion–exclusion answers the stated bad properties, not a substituted graph property.

Intersection sums and exact-hit counts

Let E_t count objects satisfying exactly t properties. An object with c hits contributes $\binom{c}{k}$ to S_k, so $S_k = \sum_{c=k}^{n} \binom{c}{k} E_c$. Define the finite polynomial $G(z) = \sum_{x \in U} z^{c(x)} = \sum_t E_t z^t$. Expand $z^{c(x)} = [1 + (z - 1)]^{c(x)}$ and interchange finite sums:

$$G(z) = \sum_{k=0}^{n} S_k (z-1)^k.$$

This polynomial identity has no convergence condition. At z=1 it gives N; at z=0 the no-hit count. Coefficient comparison gives inversion.

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Exact-hit inversion with a pointwise proof

The coefficient of z^t gives

$$E_t = \sum_{k=t}^{n} (-1)^{k-t} \binom{k}{t} S_k.$$

An object with c<t contributes zero, and one with c=t contributes one. For c>t,

$$\sum_{k=t}^{c} (-1)^{k-t} \binom{k}{t} \binom{c}{k} = \binom{c}{t} \sum_{j=0}^{c-t} (-1)^j \binom{c-t}{j} = 0.$$

Here $\binom{k}{t}\binom{c}{k} = \binom{c}{t}\binom{c-t}{k-t}$: both choose t distinguished elements and k−t extra ones. Thus precisely the target objects remain. The cases t=0 and t=n give inclusion–exclusion and E_n=S_n respectively.

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At-least-hit coefficients differ

For 1≤t≤n, let $H_t = \sum_{j=t}^{n} E_j$. Substituting inversion and swapping sums gives the coefficient of S_k as

$$\sum_{j=t}^{k} (-1)^{k-j} \binom{k}{j} = (-1)^{k-t} \binom{k-1}{t-1},$$

by the truncated binomial identity with index k−j. Therefore

$$H_t = \sum_{k=t}^{n} (-1)^{k-t} \binom{k-1}{t-1} S_k.$$

At t=1 this is the union formula. At least zero properties means N directly; do not substitute t=0 into a lower index −1. Exact and at-least formulas cannot be interchanged by merely dropping terms.

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Recovering the full three-property distribution

For $N=20, S_1=18, S_2=7, S_3=1$,

$$E_3 = 1, \quad E_2 = 7 - 3 = 4, \quad E_1 = 18 - 2 \cdot 7 + 3 \cdot 1 = 7, \quad E_0 = 8.$$

The total $8+7+4+1=20$. Reconstruct $S_1=7+2 \cdot 4+3 \cdot 1=18$ and $S_2=4+3 \cdot 1=7$. At least two gives $H_2=S_2-2S_3=5=E_2+E_3$. No hits is the union complement, size 8; the triple-intersection complement has size 19 and answers another question.

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Derangements and their formula

A permutation π:[n]→[n] is a bijection. It is a derangement if every π(i)≠i; write its count D_n. Let A_i mean i is fixed. Specifying k fixed points leaves (n−k)! permutations, so

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

The empty permutation gives $D_0=1$, while $D_1=0$. No fixed point differs from merely not being the identity. Swapping 1,2 while fixing the rest satisfies only the latter when fixed points remain.

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Two reversible classes prove the recurrence

For $n \geq 2$ fix $\pi(1)=j \neq 1$, with $n-1$ choices. If $\pi(j)=1$, delete the two-cycle $(1\ j)$, leaving an $(n-2)$-point derangement.

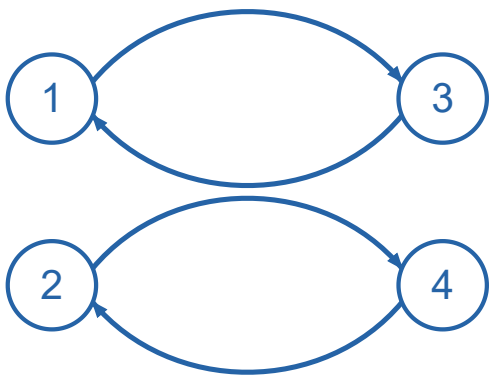
Otherwise let k be the unique predecessor of 1, so $k \neq j$. Contract $k \to 1 \to j$ to $k \to j$ and delete 1. No new fixed point occurs: the only possibility would require $k=j$, already excluded. Conversely, in any derangement of the remaining $n-1$ labels, insert 1 just before j at its unique predecessor. This uniquely restores the second class.

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Two reversible classes prove the recurrence (continued)

Fix π(1)=3: two-cycle deletion or longer-cycle contraction

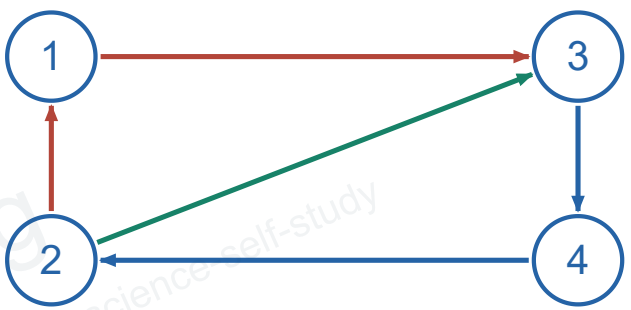
Two-cycle: remove (1 3)



Remove 1,3

Keep 2↔4

Longer cycle: contract 2→1→3 to 2→3



Remove red edges; add green. Inverse: insert 1 before 3.

The classes are disjoint and exhaustive, proving $D_n = (n-1)(D_{n-1} + D_{n-2})$. The chosen j remains the image of 1; do not silently relabel it as another point in the reduced problem.

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Integer traces for five-point derangements

$D_0=1, D_1=0$ give $D_2=1, D_3=2, D_4=3(2+1)=9, D_5=4(9+2)=44, D_6=5(44+9)=265$.

k	$\binom{5}{k}(5-k)!$	Sign
0	120	+
1	120	−
2	60	+
3	20	−
4	5	+
5	1	−

Thus $120-120+60-20+5-1=44$, matching the recurrence. A permutation with three fixed points appears in several intersection sums; these rows are not disjoint permutation classes.

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Distance from n!/e and nearest integers

The alternating remainder in $e^{-1} = \sum_{k\geq 0}(-1)^k/k!$ gives, for n≥1,

$$|D_n - n!/e| < 1/(n+1).$$

Even n lies above n!/e, odd n below. For n≥2 the error is below one half; n=1 uses 1/e<1/2 directly. Thus

$$D_n = \lfloor n!/e + 1/2 \rfloor \qquad (n \geq 1).$$

At n=0 use $D_0=1$ separately. Uniform-permutation probability D_n/n! tends to e^{-1} rather than equaling it for every finite n. Numeric rounding also needs controlled computational error; the mathematical remainder alone is no floating-point guarantee.

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Exactly fixed points and relative derangements

Choose t fixed points, then derange all others, giving $\binom{n}{t} D_{n-t}$. These disjoint exhaustive classes give $n! = \sum_t \binom{n}{t} D_{n-t}$. For n=5, counts at t=0,…,5 are 44,45,20,10,0,1, totaling 120. Exactly n−1 fixed points is impossible: bijectivity forces the last point fixed too.

For two assignment rounds with no person receiving the same item twice, fix any first-round bijection and relabel by it. The second is a relative derangement, giving n!D_n ordered pairs of rounds. It is not D_n^2 because the first round has no original-label restriction.

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Rook polynomials on a specified board

For $B \subseteq [m] \times [n]$, r_k(B) counts k rooks in B using distinct rows and columns; r_0=1 and $R_B(x) = \sum_k r_k(B) x^k$. At most min(m,n) rooks fit.

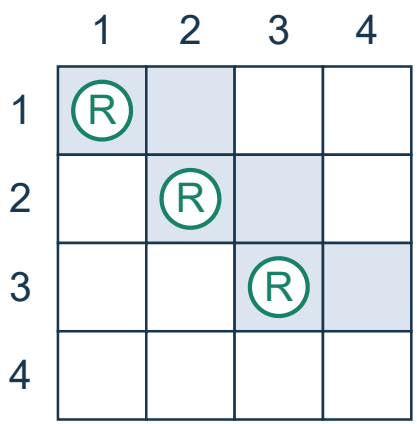
B specifies where rooks are counted. When B later represents forbidden assignment positions, rooks still lie in B to count bad-event intersections; B does not become allowed assignment cells. On a full m×n board, choose k rows, k columns, and a bijection: $r_k = \binom{m}{k}\binom{n}{k}k!$. The full 2×2 board gives $1+4x+2x^2$, not $(1+x)^4$, which would permit attacks.

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Every coefficient of the six-cell ladder

In 4×4 take $B=\{(1,1),(1,2),(2,2),(2,3),(3,3),(3,4)\}$.

Rooks on forbidden B: nonattacking cells give compatible bad events



B shaded; R: three rooks

Three-rook column triples

$r_0=1$; $r_1=6$

(1,2,3) (1,2,4)

$r_2=C(6,2)-3-2=10$

(1,3,4) (2,3,4)

$r_3=4$; $r_4=0$

At most one rook per row.

No fourth nonempty row, so four rooks are impossible.

$r_1=6$. Among $\binom{6}{2}=15$ cell pairs, three share rows and two share columns. Distinct cells cannot share both, so $r_2=10$. Three rooks require one in each of the first three rows. Their column triples are exactly (1,2,3),(1,2,4),(1,3,4),(2,3,4), giving $r_3=4$. Row four is empty, so $r_4=0$. Hence $R_B=1+6x+10x^2+4x^3$.

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Delete a cell or select it

For e=(i,j)∈B, placements either omit e and lie in $B \setminus \{e\}$, or select e and avoid its entire row and column. Call the latter remaining board B_ij. Deleting or restoring e gives reversible disjoint cases:

$$r_k(B) = r_k(B \setminus \{e\}) + r_{k-1}(B_{ij}), \quad R_B = R_{B\setminus\{e\}} + xR_{B_{ij}}.$$

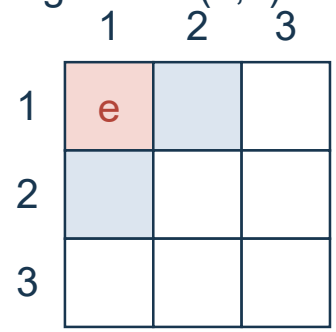
For B={(1,1),(1,2),(2,1)}, choose e=(1,1).

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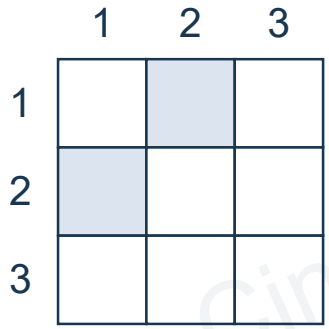
Delete a cell or select it (continued)

Deletion/contraction: disjoint cases according to whether e is used

Original: e=(1,1)

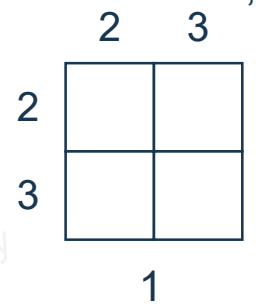


Omit e: delete one cell



$1+2x+x^2$

Select e: delete row 1, column 1



$$R_B(x) = (1+2x+x^2) + x \cdot 1 = 1+3x+x^2$$

Omitting e leaves two nonattacking cells with polynomial $(1+x)^2$. Selecting e leaves an empty board, polynomial 1. Thus $R_B=1+2x+x^2+x=1+3x+x^2$. Each branch reduces cell count; empty-board base one ensures termination.

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Board products need disjoint rows and columns

If $B=B_1 \cup B_2$ and both row sets and column sets are disjoint, a k-rook placement uniquely splits into i rooks in B_1 and k−i in B_2. Conversely two component placements cannot attack across boards. Hence

$$r_k(B) = \sum_{i=0}^{k} r_i(B_1) r_{k-i}(B_2), \qquad R_B = R_{B_1} R_{B_2}.$$

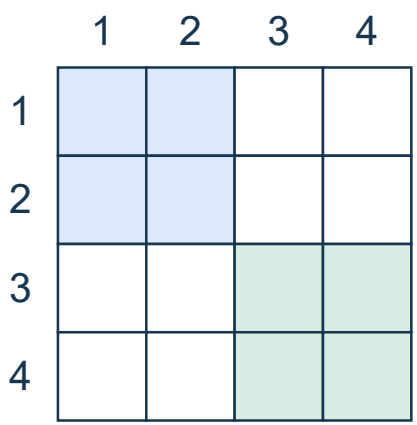
Cell disjointness alone fails. For $B_1=\{(1,1)\}, B_2=\{(1,2)\}$, each gives 1+x, but their product wrongly includes x^2; the union gives 1+2x. Splitting remains unique; combining individually legal placements fails. Row or column relabeling preserves attacks and bijects placements of each size, helping locate genuinely independent blocks.

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Full convolution for two 2×2 blocks

Take only the upper-left and lower-right full 2×2 blocks of a 4×4 board. Their rows and columns separate.

Independent rook choices require disjoint rows and columns



Blue B_1: rows 1,2 × columns 1,2

Green B_2: rows 3,4 × columns 3,4

$R_1=R_2=1+4x+2x^2$

$R_B=1+8x+20x^2+16x^3+4x^4$

$r_2=2\cdot1+4\cdot4+1\cdot2=20$

Two rooks: both in B_1, one per block, or both in B_2.

$$R_B = (1 + 4x + 2x^2)^2 = 1 + 8x + 20x^2 + 16x^3 + 4x^4.$$

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Full convolution for two 2×2 blocks (continued)

The x^2 coefficient is $2+4 \cdot 4+2=20$, for both rooks in one block or one in each. The x^3 coefficient is $4 \cdot 2+2 \cdot 4=16$. Polynomial products count independent choices and require cross terms, not just matching-degree products.

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Why forbidden-position intersections use rooks

A permutation chooses one cell in each row and column. For each forbidden (i,j)∈B, let its bad event be π(i)=j. If specified k cells share a row or column, their intersection is empty. Otherwise fixed nonattacking cells extend to (n−k)! bijections of remaining rows and columns. There are r_k(B) such cell sets, so

$$N_{\text{avoid}} = \sum_{k=0}^{n} (-1)^k r_k(B)(n-k)!.$$

This counts complete avoidance, not exactly k forbidden hits. A diagonal board gives R_B=(1+x)^n and the derangement formula. An empty board gives n!.

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All ladder-board avoiding permutations

The coefficients 1,6,10,4 at n=4 give

$$24 - 6 \cdot 6 + 10 \cdot 2 - 4 \cdot 1 = 4.$$

The row-ordered permutations are (3,1,2,4),(3,4,1,2),(3,4,2,1),(4,1,2,3). Row one permits only 3,4; row two forbids 2,3; row three forbids 3,4; each result uses every column once. Subtracting bad events independently repeats corrections to the same permutation. Rook coefficients organize jointly possible bad cells; factorials count their full extensions.

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Rectangular injections retain a denominator

Assign m named jobs injectively to n named people, m≤n. Total assignments are (n)_m=n!/(n−m)!. Fixing k nonattacking forbidden cells leaves

$$(n-k)!/(n-m)!$$

injections of the other jobs. Therefore $N_{\text{avoid}}=\sum_{k=0}^{m}(-1)^k r_k(B)(n-k)!/(n-m)!$. Attacking cells still give empty intersections. At m>n there are no injections; return zero without negative factorials. At m=0 the unique empty assignment gives one. At m=n the denominator is 0!=1, recovering the square case.

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Two jobs and four people

Take B={(1,1),(1,2),(2,2),(2,3)}. Here $r_0=1$,$r_1=4$; legal two-rook column pairs by row are (1,2),(1,3),(2,3), so $r_2=3$.

$$N = 4!/2! - 4 \cdot 3!/2! + 3 \cdot 2!/2! = 12 - 12 + 3 = 3.$$

Job one permits 3,4; job two permits 1,4. Remove the repeated-person choice (4,4), leaving (3,1), (3,4),(4,1). Omitting 2! counts each injection twice by unnecessarily ordering the two unused people.

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Rook coefficients recover exact-hit distributions

Let h_t count permutations using exactly t cells of B. Since $S_k = r_k(B)(n-k)!$, inversion gives

$$h_t = \sum_{k=t}^{n} (-1)^{k-t} \binom{k}{t} r_k(B)(n-k)!.$$

The ladder has $S_0,\dots,S_4 = 24,36,20,4,0$. Thus $h_0=4, h_1=36-40+12=8, h_2=20-12=8, h_3=4, h_4=0$, totaling 24. Also $S_1 = h_1+2h_2+3h_3=36$. The forbidden-cell count $r_1=6$ differs from the exactly-one-hit permutation count 8; these quantities have different units.

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Row-by-row rook dynamic programming

Let d_i(S) count placements in the first i rows using exactly column set S. Initially $d_0(\varnothing)=1$, others zero. For each row, omit a rook or select one available column j with $(i,j)\in B$:

```
new[...] = 0
for each subset S:
    new[S] += old[S]
    for each j with (i,j) in B and j not in S:
        new[S union {j}] += old[S]
old = new
```

Read only the preceding row's old table; in-place updates could place two rooks in one row. Every legal placement uniquely determines whether this row has a rook, its column, and the previous placement. This proves completeness without duplication. Finally $r_k = \sum_{|S|=k} d_m(S)$. There are at most mn2^n additions and O(2^n) counts in two layers. This is arithmetic-operation cost; large-integer bit cost is separate. Transpose to reduce the column dimension when helpful. This is not a polynomial-time claim for large boards.

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Sources and model boundaries

Grimaldi, Discrete and Combinatorial Mathematics, fifth edition, §§8.1–8.5 cover inclusion–exclusion, surjections, bounded solutions, number theory, graphs, circular arrangements, exact/at-least hits, derangements, rook polynomials, and forbidden positions. Pointwise truncation proofs, weight conditions, and row dynamic programming supplement these topics.

Specify the universe, labels, rotation/reflection convention, injectivity, and meaning of B before counting. Each formula uses its stated model. Finite enumeration checks examples but does not replace a general proof.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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